Livestock Sector Forecasting in Kenya: A Cohort-Based Quarterly Analysis on Population, Production, and Feed Demand

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***Abstract*—**
Kenya's livestock sector contributes 42% of agricultural GDP and 12% of national GDP, yet quantitative planning remains constrained by the absence of reproducible, data-driven analytical models. This study presents a cohort-based quarterly forecasting framework spanning nine value chains: dairy cattle, beef cattle, camels, goats, sheep, poultry, pigs, rabbits, and apiculture, initialized from the 2019 KNBS livestock census with projections from 2020 to 2041. The model applies a quarterly transition sequence covering mortality, culling, offtake, age-based maturation, and reproduction across all cohorts. Production modules translate herd and flock dynamics into species-specific output metrics, while a carrying capacity constraint — derived from satellite-based vegetation indices and adjusted by a utilization factor — regulates growth against forage availability. A feed demand module estimates dry matter requirements by species and agro-ecological zone. Projected outputs indicate a CAGR of 3.6-6% across all value chains. Beyond forecasting, the framework exposes productivity bottlenecks and structural inefficiencies, including feed deficits, low offtake rates, and suboptimal reproductive performance, that aggregate assessments typically obscure. The model provides an analytical basis for food security planning, targeted infrastructure investment, and evidence-based livestock policy.

***Index Terms*—Livestock model forecasting, carrying capacity, biological parameters**

I. Introduction

Write introduction here.

This paper [1] shows an example of how citations work.

II. Related Work

Write related work (literature review) here.

III. Methodology

A. Overview

Eight value chains (both ruminant and non-ruminant animals) were selected on the basis of their contribution to Kenya's agricultural GDP.[1] For each chain, forecasting covers three domains: **population dynamics**, **production volumes**, and **feed demand**. The baseline year is 2019; the projection horizon runs from 2020 to 2041, with 2020–2025 retained for model validation against observed values.

A first-principles modelling approach was adopted. This choice reflects (i) the transparency and auditability it affords where each intermediate quantity is directly traceable to its inputs, and (ii) the heterogeneous, multi-source nature of the available data, which precludes the large homogeneous datasets required by machine-learning methods.

Parameters were compiled from peer-reviewed literature, government publications, and sector reports, then validated by livestock experts from the Ministry, associations, and farmers. Accepted values are listed in Appendix 1.[2]

The model operates on a quarterly time step. For each quarter, the three domains are computed in sequence.

B. Population Model Equations

Population is tracked across a defined cohort structure specific to each value chain. Within each quarter, the following processes are applied in order where relevant: mortality, culling, offtake, transition to the next age cohort, and births. The starting population for each cohort is updated by sequentially applying the corresponding rates, with inflows from younger cohorts and new births added at the end of the period.

1) Beef Cattle:

$$P(t+1) = \left[P(t)\,(1-m_q)(1-k_q)(1-o_q)\right]+G(t)-E(t)+B(t) \tag{1}$$

[1] Citation required, get from Natasha

[2] Summarize parameters

$P(t)$	Starting pop.	m_q	Mortality rate (qtr.)
$P(t+1)$	Ending pop.	k_q	Culling rate
$G(t)$	Inflow (young cohort)	o_q	Offtake rate (steers)
$E(t)$	Outflow to next cohort	$B(t)$	New births (calves)

2) Goats:

$$P(t+1) = \left[P(t)\,(1-m_q)(1-k_q)\right] + G(t) - E(t) + B(t) \quad (2)$$

Variable definitions follow the same convention as beef cattle; the offtake term o_q does not apply.

3) Camels:

$$P(t+1) = P(t) - D(t) - C(t) - O(t) + B(t) \quad (3)$$

$D(t)$	Natural mortality	$C(t)$	Culling offtake
$O(t)$	Commercial offtake	$B(t)$	Births in period t

4) Poultry: Three sub-sectors are modelled separately. T_{bro} denotes annual broiler throughput and S_r the rearing survival rate.

$$P_{\text{ind}}(t+1) = P_{\text{ind}}(t) - E_{\text{ind}}(t) + R_{\text{ind}}(t) + A(t) \quad (4)$$

$$P_{\text{lay}}(t+1) = P_{\text{lay}}(t) - E_{\text{lay}}(t) + \text{DOC}(t) \times S_r \quad (5)$$

$$T_{\text{bro}}(t+1) = T_{\text{bro}}(t) + A(t) \quad (6)$$

The sector addition term $A(t)$ and its sigmoid growth modifier $g(t)$ are:

$$A(t) = P(t)\,(r_{\text{ent}} + r_{\text{exp}})\,g(t) \quad (7)$$

$$g(t) = g_{\min} + (g_{\max} - g_{\min})\,\exp\left(-\kappa^2\,\frac{(t-t_0)^2}{2}\right) \quad (8)$$

5) Apiculture:

$$H(t+1) = H(t)\,(1-\delta(t))(1-\rho(t)) + S(t) \quad (9)$$

$\delta(t)$	Seasonal colony loss	$\rho(t)$	Queen replacement rate
$S(t)$	New colonies (swarming)		

6) Pigs:

$$P(t+1) = P(t) - D(t) - C(t) - O(t) + B(t) + R(t) \quad (10)$$

where $R(t)$ represents replacement additions to the breeding herd. Variable definitions for $D(t)$, $C(t)$, $O(t)$, $B(t)$ follow the camel convention.

7) Rabbits: Rabbits follow the same population equation as pigs (including $R(t)$).

C. Production Model Equations

Production volumes are derived from the population state at the end of each quarter. Each value chain has a distinct production equation reflecting the commodity it yields, that is meat, milk, eggs, honey, or wax, as detailed in Sections II-B through II-C. [3] For ruminants, production is computed separately for each cohort and then aggregated. For poultry, the three sub-sectors are modelled independently. Apiculture production is calculated at the hive level and then scaled by the number of hives.

[3]Update the section names

1) Beef Cattle:

$$\text{Beef (kg)} = \sum_c N_c \times \text{LW}_c \times \text{DP}_c \quad (11)$$

where $c \in \{\text{steers, cull cows, cull bulls}\}$, LW is live weight (kg) and DP is dressing percentage.

2) Goats:

$$\text{Meat (kg)} = (N_{\text{removed}} + N_{\text{excess bucks}}) \times \text{LW} \times \text{DP} \quad (12)$$

$$\text{Milk (kg)} = N_{\text{does}} \times f_L \times Y_d \times D_L \quad (13)$$

N_{does}	Surviving does
f_L, Y_d, D_L	Lactating fraction; daily yield (kg); lactation days
$N_{\text{excess bucks}}$	Bucks above 4% herd cap, removed for slaughter

3) Camels:

$$\text{Milk (kg yr}^{-1}) = N_{\text{adult females}} \times f_L \times Y_d \times 365 \quad (14)$$

$$\text{Meat (kg)} = \sum_c N_c \times \text{LW}_c \times \text{DP}_c \quad (15)$$

where $c \in \{\text{juv. offtake, adult offtake, cull adults}\}$.

4) Poultry:

$$\text{Eggs}_{\text{ind}}(t) = P(t) \times f_{\text{fem}} \times Y_e \quad (16)$$

$$\text{Eggs}_{\text{lay}}(t) = P(t) \times \eta_{\text{HD}} \times 365 \quad (17)$$

$$\text{Meat (kg)} = T_{\text{bro}} \times \text{LW} \times \text{DP} \quad (18)$$

where f_{fem} is female share, Y_e is eggs per hen per year, and η_{HD} is hen-day production rate.

5) Apiculture:

$$\text{Honey}_{h,q} = H_h \cdot \frac{Y_h}{4} \cdot \bar{s} \cdot (1-\varphi) \cdot \Omega(t) \cdot \varepsilon_h \quad (19)$$

$$\text{Wax}_{h,q} = \text{Honey}_{h,q} \times \omega_h \quad (20)$$

$$\text{Nectar}_{h,q} = H_h \cdot \frac{N_h}{4} \cdot \bar{s} \cdot \phi_q \cdot \nu_q \cdot \Omega(t) \quad (21)$$

Y_h, $\bar{s}$	Annual yield/hive (kg); effective colony strength
φ	Forage competition penalty (0–0.20)
ε_h	Hive-type efficiency factor
$\Omega(t)$	Technology/training improvement (sigmoid)
ω_h	Wax-to-honey ratio
ϕ_q, ν_q	Quarterly forage and nectar seasonality factors

6) Pigs and Rabbits:

$$\text{Meat} = N_{\text{slaughtered}} \times \text{LW} \times \text{DP} \quad (22)$$

D. Feed Model Equations

Feed demand is estimated using dry matter intake (DMI) for all value chains. A DMI-based approach was adopted due to limited availability of detailed ration composition data across most value chains; it provides a consistent and tractable basis for estimating feed requirements from population and live weight alone. The specific formulations vary by value chain and are presented in Section II-D.

1) Beef Cattle, Pigs, and Rabbits: All three value chains use the same daily dry matter intake (DMI) formulation:

$$\text{DMI} = P(t) \times \text{LW} \times r_{\text{DMI}} \times 365 \quad (23)$$

where r_{DMI} is daily dry matter intake as a proportion of body weight.

2) Goats: Feed is summed across cohorts (kids, weaners, growers, does, bucks):

$$\text{Feed}(t) = \sum_{\text{cohort}} N_c \times \text{BW}_c \times r_{\text{DMI},c} \times D \tag{24}$$

where D is the number of days in the period.

3) Camels: Feed is accumulated quarterly, with intake weighted by average live weight over the quarter:

$$\text{Annual feed} = \sum_{q=1}^{4} \text{Feed}_q \tag{25}$$

$$\text{Feed}_q = \sum_{\text{cohort}} N_c \times \left(\frac{\text{BW}_{0,c} + \text{ADG}_c \times 90}{2} \right) \times r_{\text{DMI},c} \times 90 \tag{26}$$

where ADG is average daily gain ($\text{kg}\,\text{d}^{-1}$).

4) Poultry: Each sub-sector has a distinct feed equation reflecting its production cycle structure:

$$\begin{aligned} F_{\text{ind}}(t) = \big[& R(t)(f_{\text{chk}}\, d_{\text{chk}} + f_{\text{gr}}\, d_{\text{gr}}) \\ & + P(t)\, f_{\text{ad}}\, 365 \big] (1-\sigma)^t \end{aligned} \tag{27}$$

$$\begin{aligned} F_{\text{lay}}(t) = \text{DOC}(t) & (f_{\text{chk}}\, d_{\text{chk}} + f_{\text{pul}}\, d_{\text{pul}}) \\ & + P(t)\, f_{\text{hen}}\, 365 \end{aligned} \tag{28}$$

$$F_{\text{bro}} = N_{\text{birds}} \sum_{\text{phase}} f_{\text{phase}} \times d_{\text{phase}} \times C_{\text{yr}} \tag{29}$$

where $f_{(\cdot)}$ is phase-specific intake ($\text{kg}\,\text{d}^{-1}$), $d_{(\cdot)}$ is phase duration (days), σ is the scavenging offset, and C_{yr} is production cycles per year.

IV. Results

All projections run from a 2019 baseline to 2041, with 2020–2025 serving as the validation window (shaded in all figures). Compound annual growth rates (CAGRs) are reported over the full 2019–2041 horizon unless noted otherwise.

A. Ruminant Value Chains

1) Population: Cattle dominate the ruminant sector by absolute numbers. Beef cattle grow from 16.6 million in 2019 to 67.8 million by 2041 (CAGR 6.59%), while dairy cattle expand from 4.5 million to 13.1 million (CAGR 4.93%). Goats are the most numerous ruminant throughout the horizon, rising from 34.7 million to 121.3 million (CAGR 5.86%). Camels grow steadily from 4.7 million to 14.3 million (CAGR 5.15%), and sheep show the most modest growth, from 27.4 million to 55.7 million (CAGR 3.27%). Population trajectories are shown in Fig. 1.

2) Production: Milk production is led by dairy cattle (3.1 billion kg in 2019 to 9.3 billion kg by 2041; CAGR 5.12%), followed by camels (0.99 billion kg to 5.6 billion kg; CAGR 8.23%) and goats (156 million kg to 531 million kg; CAGR 5.71%). Chevon (goat meat) records the highest CAGR among all ruminant products at 17.02%, driven by the large baseline herd adjustment in 2020. Beef production grows from 322 million kg to 1.37 billion kg (CAGR 6.82%), and mutton from 49 million kg to 165 million kg (CAGR 5.65%). Wool production (sheep) is modest

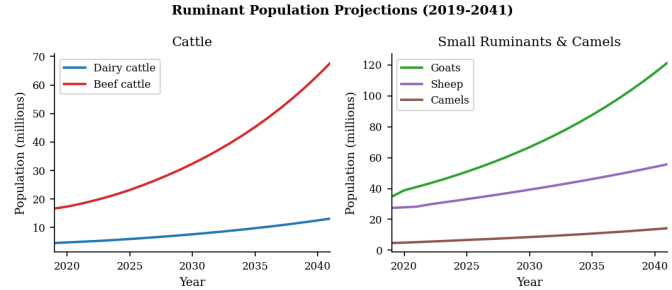

Fig. 1: Ruminant population projections, 2019–2041. Shaded region indicates the 2020–2025 validation window.

but grows consistently (CAGR 7.25%). Production trends are presented in Fig. 2.

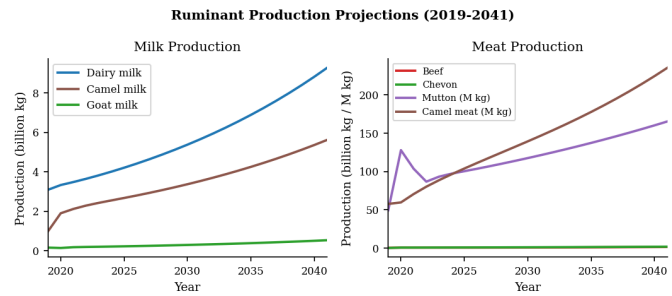

Fig. 2: Ruminant production projections, 2019–2041.

3) Feed Demand: Beef cattle are the dominant feed consumer throughout the horizon, rising from 57.4 million tonnes DM to 218.3 million tonnes DM (CAGR 6.26%). Dairy cattle demand grows from 14.3 million to 41.2 million tonnes DM (CAGR 4.92%). Camel feed demand expands at CAGR 6.66%, from 10.2 million to 42.2 million tonnes DM. Goat and sheep feed series begin in 2020 (no 2019 parameter baseline) and grow at CAGR 6.09% and 3.33% respectively over their available range. Feed demand trajectories are shown in Fig. 3.

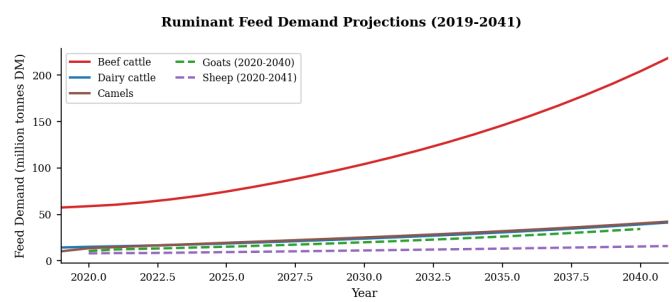

Fig. 3: Ruminant feed demand projections, 2019–2041. Goat and sheep series begin in 2020; the goat 2041 value is excluded due to a data entry anomaly.

B. Non-Ruminant Value Chains

1) Population: Indigenous poultry is the largest non-ruminant population, growing from 46.3 million to 89.3 million birds (CAGR 3.03%). Broiler throughput and layer flock size are relatively stable, growing at CAGR 0.43% and 0.42% respectively, reflecting sector saturation assumptions. Pig and rabbit populations expand more rapidly: pigs from 596 thousand to 2.09 million (CAGR 5.86%) and rabbits from 737 thousand to 3.24 million (CAGR 6.95%). Total apiculture hive count grows from 1.29 million to 3.49 million across all four hive types (log, KTBH, Langstroth, box). Population trends are shown in Fig. 4.

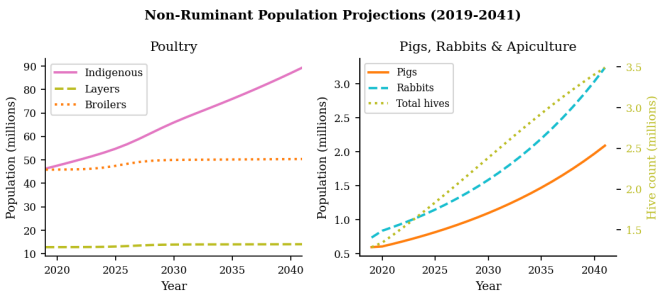

Fig. 4: Non-ruminant population projections, 2019–2041. Right panel: pig and rabbit populations (left axis) and total hive count (right axis).

2) Production: Poultry meat production grows modestly from 82.4 million kg to 109.7 million kg (CAGR 1.31%), consistent with the near-stable flock projections. Egg production (layers and indigenous combined) is reported in trays from 2020 onward — the 2019 value is excluded as it appears to reflect a different unit convention — and grows from 180.4 million trays to 228.7 million trays (CAGR 1.14%, 2020–2041). Pork and rabbit meat grow in parallel at approximately CAGR 4.0%. Honey production expands from 13.9 million kg to 41.6 million kg (CAGR 5.12%) and beeswax from 1.4 million kg to 6.5 million kg (CAGR 7.35%). Production trends are shown in Fig. 5.

3) Feed Demand: Poultry feed demand is the largest non-ruminant feed consumer, growing from 1.87 million tonnes to 2.83 million tonnes DM (CAGR 1.89%). Pig feed demand grows from 448 thousand tonnes to 1.09 million tonnes DM (CAGR 4.13%). Rabbit feed demand, though small in absolute terms (18 thousand to 53 thousand tonnes DM), grows at CAGR 4.94%. Apiculture nectar demand is estimated at 852 thousand tonnes in 2019 rising to 2.49 million tonnes by 2041 (CAGR 5.00%). Feed demand trajectories are shown in Fig. 6.

C. CAGR Summary

Table I summarises the CAGRs for all value chains across all three domains. All figures are 2019–2041 unless noted.

V. Conclusion

Write conclusion here.

Fig. 5: Non-ruminant production projections, 2019–2041. Egg production plotted from 2020 onward only.

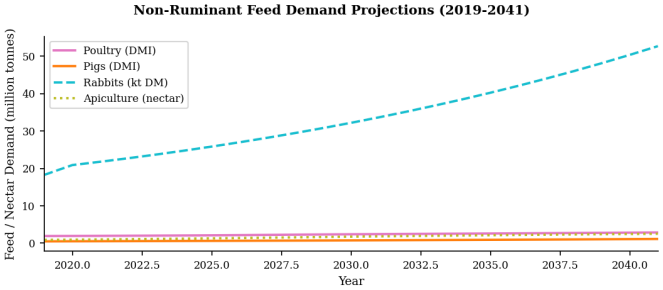

Fig. 6: Non-ruminant feed demand projections, 2019–2041. Apiculture values represent nectar demand rather than DMI.

Appendix

References

[1] S. T. Bahta, F. N. Wanyoike, L. Kirui, C. Mensah, D. K. Enahoro, J. T. Karugia, and I. Baltenweck, "Livestock sector transformation in Kenya: Current state and projections for the future," Dec. 2023.

TABLE I: Compound Annual Growth Rates (2019–2041)

Value Chain	**Indicator**	**CAGR (%)**
Ruminants — Population		
Dairy cattle	Total population	4.93
Beef cattle	Total population	6.59
Goats	Total population	5.86
Sheep	Total population	3.27
Camels	Total population	5.15
Ruminants — Production		
Dairy cattle	Milk (kg)	5.12
Beef cattle	Beef (kg)	6.82
Goats	Milk (kg)	5.71
Goats	Chevon (kg)	17.02
Sheep	Mutton (kg)	5.65
Sheep	Wool (kg)	7.25
Camels	Milk (kg)	8.23
Camels	Meat (kg)	6.61
Ruminants — Feed Demand		
Dairy cattle	(t DM)	4.92
Beef cattle	(t DM)	6.26
Goats	(t DM, 2020–2040)	6.09
Sheep	(t DM, 2020–2041)	3.33
Camels	(t DM)	6.66
Non-Ruminants — Population		
Poultry (indigenous)	Birds	3.03
Poultry (layers)	Birds	0.42
Poultry (broilers)	Throughput	0.43
Pigs	Total	5.86
Rabbits	Total	6.95
Non-Ruminants — Production		
Poultry	Meat (kg)	1.31
Poultry	Eggs (trays, 2020–2041)	1.14
Pigs	Pork (kg)	3.99
Rabbits	Meat (kg)	4.01
Apiculture	Honey (kg)	5.12
Apiculture	Wax (kg)	7.35
Non-Ruminants — Feed Demand		
Poultry	(t DM)	1.89
Pigs	(t DM)	4.13
Rabbits	(t DM)	4.94
Apiculture	Nectar (t)	5.00